

Laboratory Work 1 International Airport Database, ER Diagram

Course: Databases

Topic: International Airport Database

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Introduction

The purpose of this laboratory work is to design an Entity-Relationship Diagram for an International Airport database. The system is designed to manage information about airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks and security checks.

REQUIRED TASKS:

TASK 1: Identified entities and brief description

1. AIRPORT: Stores each airport and its location.
2. AIRLINE: Stores airline companies operating flights.
3. FLIGHT: Stores schedule, gates, actual times, airline, and both endpoint airports.
4. PASSENGER: Stores personal and passport information.
5. BOOKING: Connects one passenger to one flight and stores details.
6. BOOKING_CHANGE: Keeps the history of changes to a booking.
7. BOARDING_PASS: Stores the boarding info generated for a booking.
8. BAGGAGE: Stores individual registered baggage items
9. BAGGAGE_CHECK: Stores baggage check records
10. SECURITY_CHECK: Stores baggage check records

TASK 2: Determine attributes

AIRPORT

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	airport_id	INTEGER	unique airport identifier
	airport_name	VARCHAR(100)	name
	country	VARCHAR(100)	country
	state	VARCHAR(100)	region
	city	VARCHAR(100)	city
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

AIRLINE

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	airline_id	INTEGER	unique airline identifier
UQ	airline_code	INTEGER	code
	name	VARCHAR(100)	company name
	country	VARCHAR(100)	country
	created_at	TIMESTAMPTZ	creation time

	updated_at	TIMESTAMPTZ	last changed time
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FLIGHT

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	flight_id	INTEGER	Unique scheduled flight id
FK	airline_id	INTEGER	references
FK	departure_airport_id	INTEGER	references
FK	arrival_airport_id	INTEGER	references
	departing_gate	VARCHAR(100)	Departure gate
	arriving_gate	VARCHAR(100)	arrival gate
	scheduled_departure	TIMESTAMPTZ	planned departure time
	scheduled_arrival	TIMESTAMPTZ	planned arrival time
	actual_departure	TIMESTAMPTZ	actual departure time
	actual_arrival	TIMESTAMPTZ	actual arrival time

	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

PASSENGER

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	passenger_id	INTEGER	Unique passenger identifier
	first_name	VARCHAR(100)	first name
	last_name	VARCHAR(100)	last name
	gender	VARCHAR(100)	gender
	date_of_birth	DATE	birth date
	citizenship_country	VARCHAR(100)	citizenship
	residence_country	VARCHAR(100)	country lives
UQ	passport_number	VARCHAR(100)	passport number
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

BOOKING

KEY	ATTRIBUTE	DATA TYPE	MEANING
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PK	booking_id	INTEGER	Unique booking identifier
FK	flight_id	INTEGER	References
FK	passenger_id	INTEGER	References
	status	VARCHAR(100)	Booking state
	booking_platform	VARCHAR(100)	platform made booking
	ticket_price	NUMERIC(12,2)	booking price
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

BOOKING_CHANGE

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	change_id	INTEGER	Unique change identifier
FK	booking_id	INTEGER	References
FK	old_flight_id	INTEGER	References
FK	new_flight_id	INTEGER	References
	changed_time	TIMESTAMPTZ	Time the change occurred

BOARDING_PASS

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	boarding_pass_id	INTEGER	Unique boarding pass identifier
FK	boarding_id	INTEGER	References
	seat	VARCHAR(100)	Assigned seat
	boarding_time	TIMESTAMPTZ	time
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

BAGGAGE

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	baggage_id	INTEGER	Unique baggage item identifier
FK	booking_id	INTEGER	References
	weight_in_kg	NUMERIC(6,2)	Item weight in kilograms
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

SECURITY_CHECK

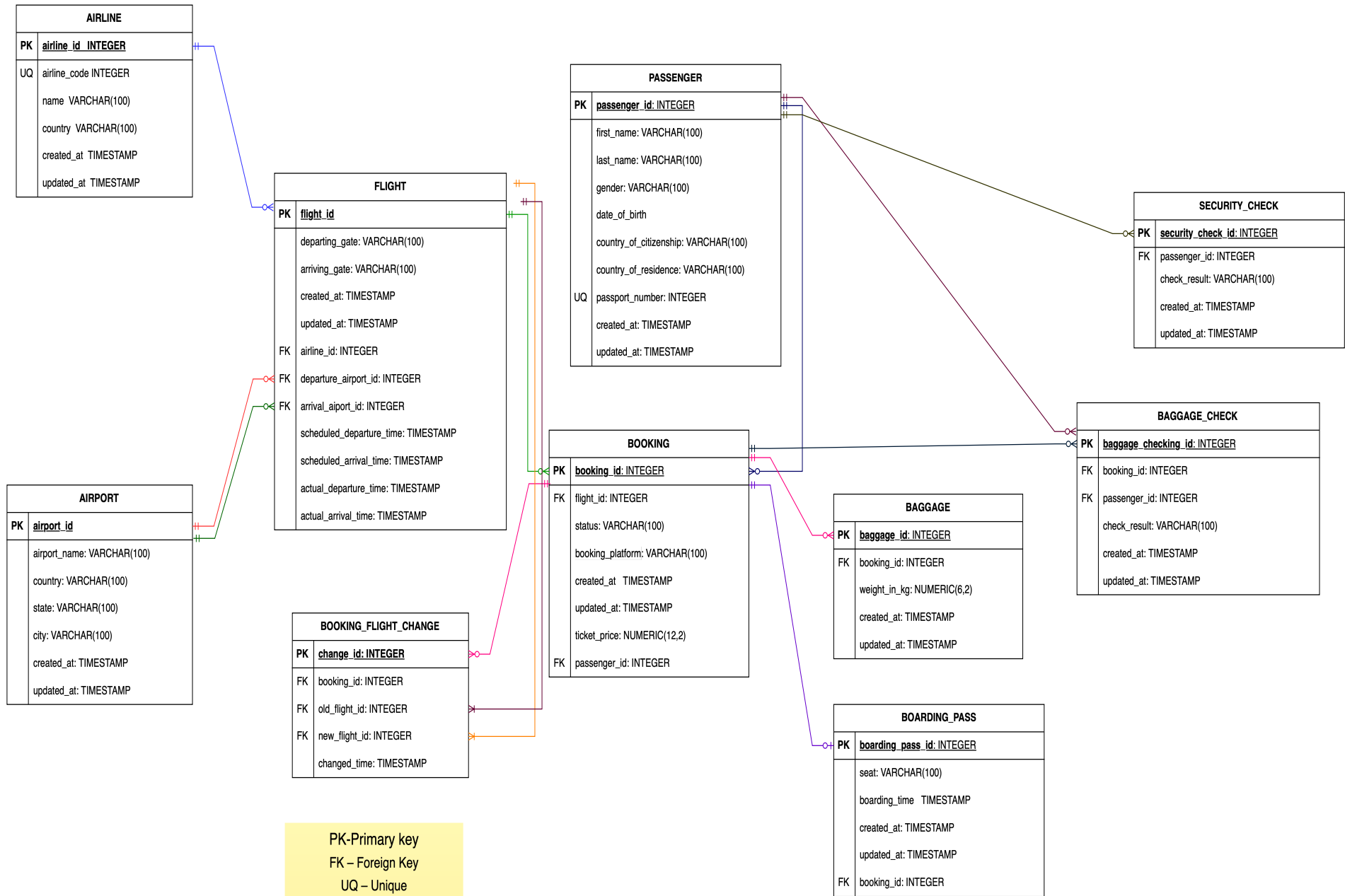
KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	security_check_id	INTEGER	Unique security check identifier
FK	passenger_id	INTEGER	References
	check_result	VARCHAR(100)	check outcome

	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

BAGGAGE_CHECK

KEY	ATTRIBUTE	DATA TYPE	MEANING
PK	baggage_checking_id	INTEGER	Unique baggage check identifier
FK	booking_id	INTEGER	References
FK	passenger_id	INTEGER	References
	check_result	VARCHAR(100)	check outcome
	created_at	TIMESTAMPTZ	creation time
	updated_at	TIMESTAMPTZ	last changed time

Entity Relational Diagram



PK-Primary key
FK – Foreign Key
UQ – Unique
I = one
O = optional / zero
crow's foot = many

TASK : Identify relationships

Cardinality and Participation Constraints

1 = exactly one ,mandatory

0/1 = zero or one ,optional

1/N = one or many,mandatory

0/N = zero or many,optional

AIRLINE-operates-FLIGHT: 1:0/N **Airline may have no/many flights; Flight must have one Airline**

AIRPORT-of departure-FLIGHT(DEPARTURE): 1:0/N **Airport may be departure point for no/many flights; Flight must have one departure airport**

AIRPORT-of arrival-FLIGHT(ARRIVAL): 1:0/N **Airport may be arrival point for no/many flights; Flight must have one arrival airport**

FLIGHT-has-BOOKING: 1:0/N **Flight may have no/many bookings; Booking must refer to one flight**

PASSENGER-make-BOOKING: 1:0/N **Passenger may have no/many bookings; Booking must belong to one passenger**

BOOKING-register-BAGGAGE: 1:0/N **Booking may have no/many bags; Baggage must belong to one booking**

BOOKING-produce-BOARDING_PASS: 1:0/1 **Booking may have no boarding pass yet; Boarding Pass must belong to one booking**

PASSENGER-passes-SECURITY_CHECK: 1:0/N **Passenger may have multiple checks; each check belongs to one passenger**

BOOKING-has-BAGGAGE_CHECK:1:0/N **Booking may have multiple baggage checks**

PASSENGER-has-BAGGAGE_CHECK: 1:0/N **Passenger may be associated with multiple baggage checks**

BOOKING-records old/new-BOOKING_FLIGHT_CHANGE: 1:0/N **Booking may have multiple changes;each change belongs to one booking.**

Conclusion

In this laboratory work, an ERD for an International Airport database was designed. The main entities, their attributes, primary keys and foreign keys were identified.